GEK-83830

DRIVE SYSTEM INSTRUCTIONS

TECHNICAL MANUAL
for
BOSCHERT
TWO-STAGE POWER SUPPLY

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

****
* NOTICE *
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This manual duplicates, with permission, the Boschert Two-Stage Power Supply Maintenance Manual. If additional information is desired or should particular problems arise which are not covered sufficiently for the customer's purposes, the matter should be referred to the following address:

Boschert, Incorporated
384 Santa Trinita Avenue
Sunnyvale, CA 94086

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TWO STAGE (PUSH-PULL OR HALF-BRIDGE) POWER SUPPLY

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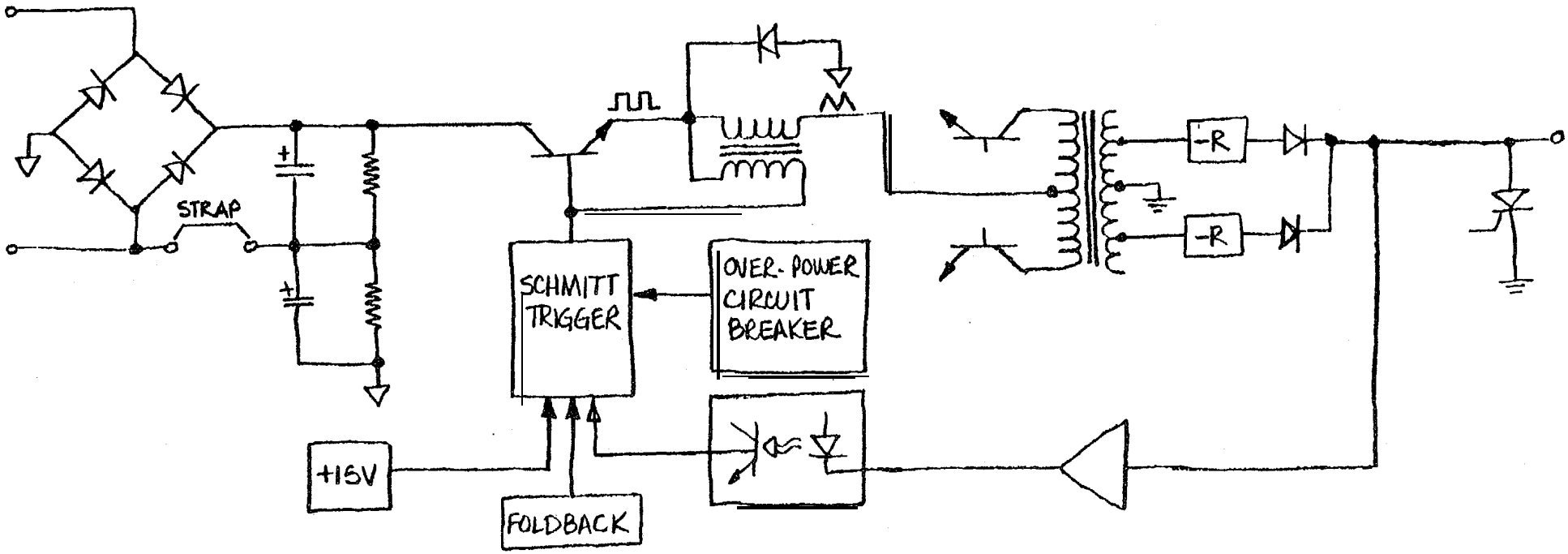
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NOTE: The descriptions and component designations in this manual refer to the OL400-3015. However, all push-pull supplies have analogous circuits which operate in a manner similar to the OL400-3015.

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FIGURE 1

TWO STAGE (PUSH-PULL OR HALF BRIDGE) BLOCK DIAGRAM



INPUT CIRCUIT

VOLTAGE DOUBLER

+15V REGULATOR

POWER SWITCH AND DRIVERS

POWER FOLDBACK

FAST TURN-OFF CIRCUIT

OVER POWER CIRCUIT BREAKER

OPTO-ISOLATOR

20kHz SATURATING REACTOR INVERTER (PUSH-PULL SHOWN)

ERROR AMPLIFIER

NEGATIVE RESISTOR

OUTPUT OVERVOLTAGE CROWBAR

TWO STAGE (PUSH-PULL OR HALF-BRIDGE) POWER SUPPLY

GENERAL THEORY OF OPERATION

Refer to the Block Diagram, Figure 1. The AC line passes through an RFI filter intended to keep switcher noise from being transmitted back out the AC line. The input voltage is rectified and the resulting DC energy is stored in the input capacitor(s). These capacitors provide both filtering the the pulsating DC from the rectifier, and "holdup," or a short term energy source, for the supply in the event of line voltage failure.

The transistor switch chops the DC input from the capacitor into a pulse train. The pulse width is modulated by the feedback loop to maintain one output at a constant voltage. The pulse train is then filtered by the primary inductor to provide a smooth DC output. The primary inductor is the main energy storage element in this two-stage design.

The filtered DC voltage from the primary inductor drives the inverter. Boschert uses two types of inverters, push-pull and half bridge, which use a saturating reactor to determine their frequency, about 20kHz. Each type is used for different manufacturability reasons. However, their operation and purpose is virtually the same. The free-running inverter transforms the high DC input voltage to the various output voltages via the output transformer.

The outputs of the transformer are full-wave rectified and filtered using LC circuits. This filtering is simple, cost effective, yet the output is regulated well enough for virtually all digitally based applications and many analog applications. In most cases all outputs share a common return.

A feedback loop, which uses an opto-isolator to isolate the primary (which floats at the line potential) and secondary, regulates one output. Almost without exception the +5V output is regulated.

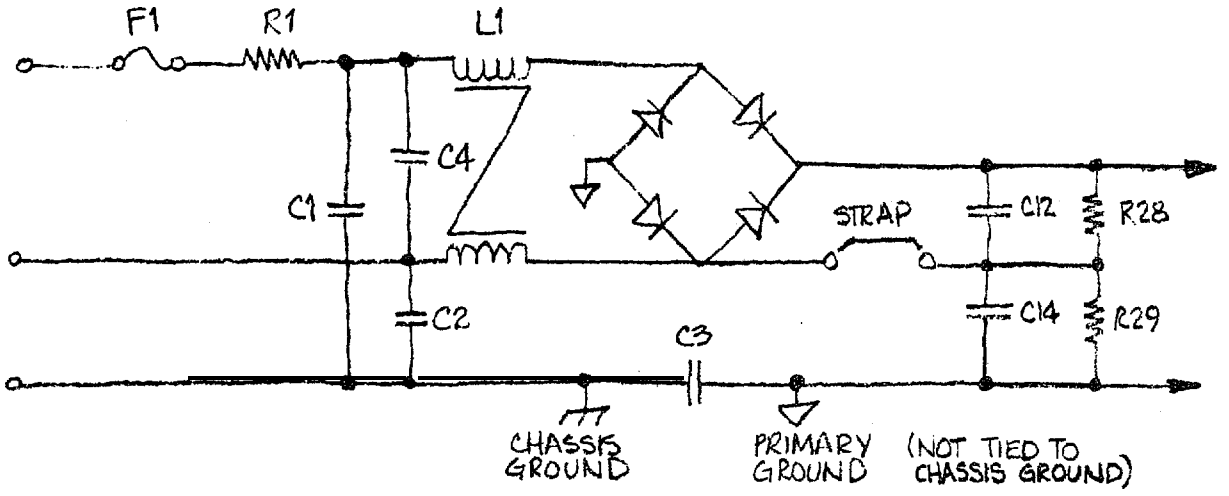


Figure 2

Input Ciruit

INPUT CICUITRY

a) RFI Filter consists of L1, Cl, C2, and C4. The purpose of this circuit is to filter out 20kHz switching noise, preventing it from being transmitted back out the input line.

b) Fuse Fl is included to protect the p.c. board traces and to reduce fire and personal hazard in the event of catastrophic supply failure. Boschert strongly recommends the use of an additional, external line fuse for further protection. These fuses are recommended:

Model	Littlefuse 3AG Series or Equivalent
OL130 thru OL153	3A
OL200 thru OL400	6-1/4A

Table 1

Line Fuse Requirements

c) Voltage Doubler Circuit is a method of allowing the supply to operate from either a 115VAC or 220VAC source. In the 115VAC mode (strap in), capacitors C12 and C14 charge on alternate line half cycles. The voltage which appears across each capacitor is approximately the peak input voltage (about 150V). Thus, the total voltage to the switching regulator is about 300V.

In the 220VAC mode (strap out), the input rectifiers act as a full wave bridge rectifier, charge C12 and C14 to approximately the peak input voltage (about 300V). Resistors R28 and R29 force C12 and C14 to charge equally, and discharge the capacitors when the power is turned off.

d) Inrush Limiting is accomplished with thermistor R1. When cool, its resistance is high. When the supply is initially turned on, it prevents a huge surge current from flowing into C12 and C14, which are initially discharged. In operation, the normal input current quickly heats R1, decreasing its resistance by a factor of about 10. Since C12 and C14 are fully charged by this time, there is no further need for surge limiting. When the supply is turned off, R1 cools and C12 and C14 discharge. The circuit is designed so that the discharge time constant and cooling time constant are roughly equal. Thus, if the supply is turned back on before R1 has fully cooled, the inrush current is limited by a combination of the charge remaining on C12 and C14, and the resistance of R1.

e) Energy storage of capacitors C12 and C14 insure that the supply will continue to operate within regulation limits for a minimum of 16ms after the input line has fallen below its limits or failed. This allows time for information in, say, a volatile memory to be stored permanently before system failure following line interruption or line brownout.

+15V REGULATOR

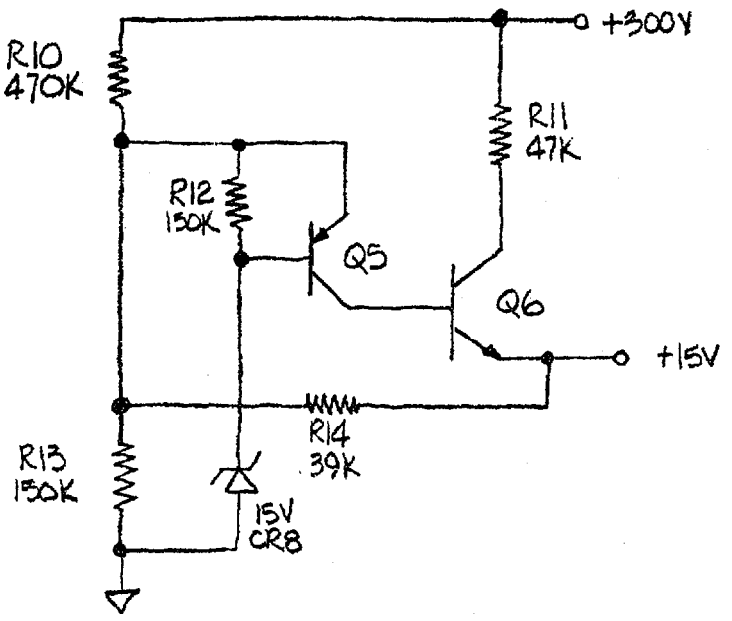


Figure 3
+15V Regulator Circuit

This circuit provides a stable reference for the switching regulator drive circuit and the 723 integrated circuit. The regulator operates as a Schmitt trigger. That is, it snaps on when the input voltage gets to approximately 75V, and snaps off when the input voltage falls below approximately 60V. This means that the power to the 723, and hence the entire supply, is either on or off. There is no "grey" area.

The circuit begins to operate when CR8 starts to conduct. This saturates Q6 which drives emitter follower Q5. Positive feedback from R14 provides hysterisis allowing the regulator to "snap" on. Current thru R14 tries to force the emitter of Q6 higher. Since the base of Q6 is at a fixed voltage, Q6 conducts harder, which in turn drives Q5. The circuit stabilizes when the emitter of Q6 is about +15V.

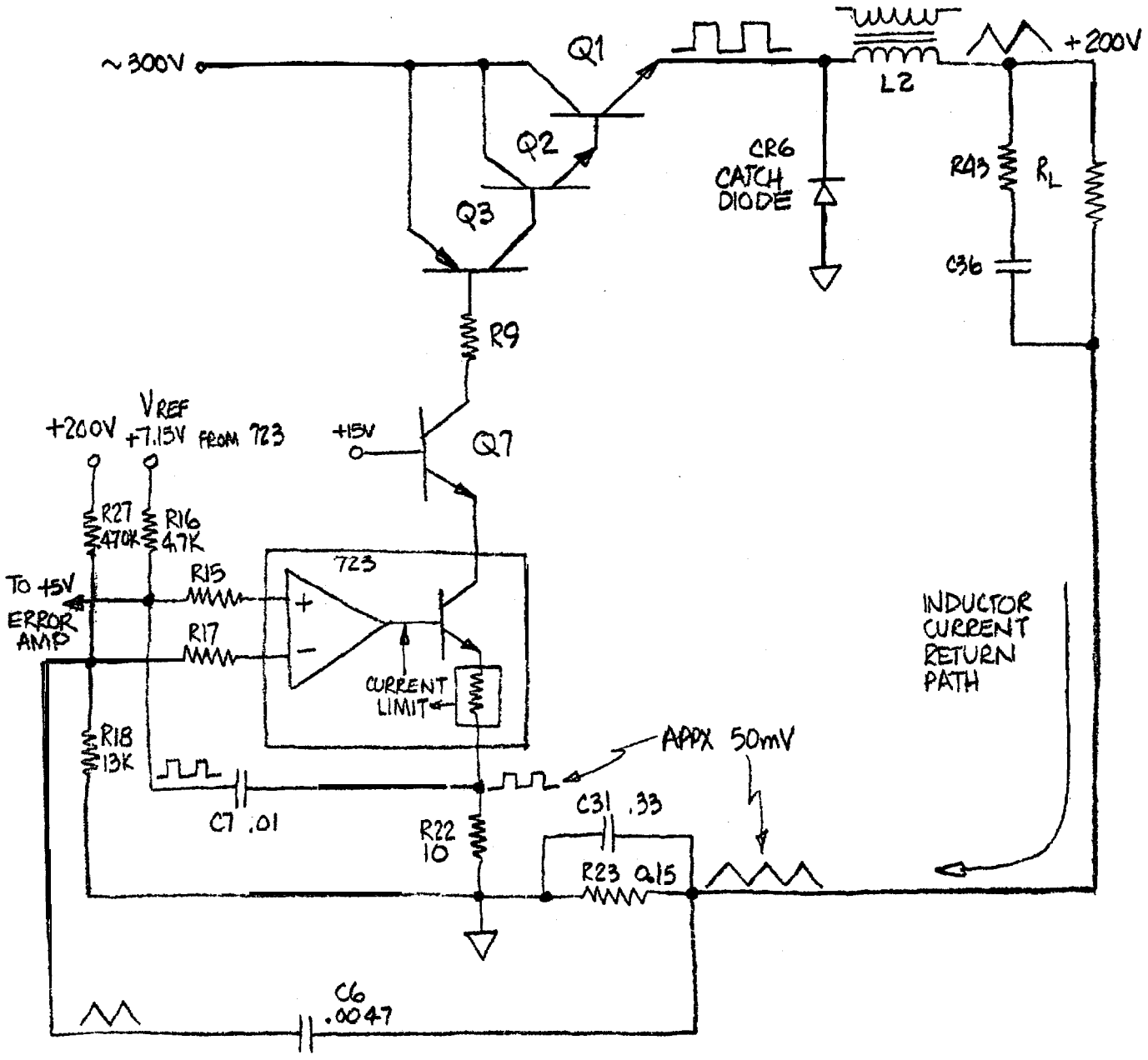


FIGURE 4
POWER SWITCH CIRCUIT

POWER SWITCH AND DRIVERS

The modulated power switch Q1 provides a chopped DC output to the primary inductor. The circuit driving Q1 consists of a 723 integrated circuit and associated current buffers (Q2 and Q3) and isolator (Q7). Notice in Figure 4 that the inverter and output loads have been represented by R_L for simplicity. Also note that the 723 , drivers, switch and feedback loop form a free running multivibrator that will continue to run even if the main (+5V) feedback loop is broken (which is shown in Figure 4 as "to 5V error amp").

The 723 is the basic switching control circuit. Although intended to be used in a linear mode, it is used in a switching mode here. The non-inverting input is referenced to about +5V. This voltage level is modulated by the +5V feedback circuit. In addition, a 50mV square wave hysterisis signal also appears on this input.

The inverting input is also referenced to about +5V. In addition, a 50mV triangle wave appears at this input. The triangle wave is generated by the primary inductor current ramping up and down linearly in response to a step voltage change on its input. This ramping current flowing thru R23 generates the triangle wave. The signal is fed to the 723 via C6, and is compared to the voltage on the non-inverting input. When the current (and hence voltage on the inverting input) reaches a predetermined value, the 723 toggles, turning off the switch Q1 and the current thru the primary inductor begins to ramp down. Since no current can flow through Q1 while it is off, the catch diode provides the necessary return current path.

By now the 50mV hysteresis square wave on the non-inverting input has disappeared by virtue of the fact that no switch current is flowing thru R22. The triangle wave on the inverting input is ramping down to the new threshold on the non-inverting input. When it reaches this point, the 723 will toggle, turning on switch Q1, the 50mV hysterisis signal will again appear on the non-inverting input, the current will begin to ramp up, and the cycle will repeat itself.

FAST TURN OFF CIRCUIT

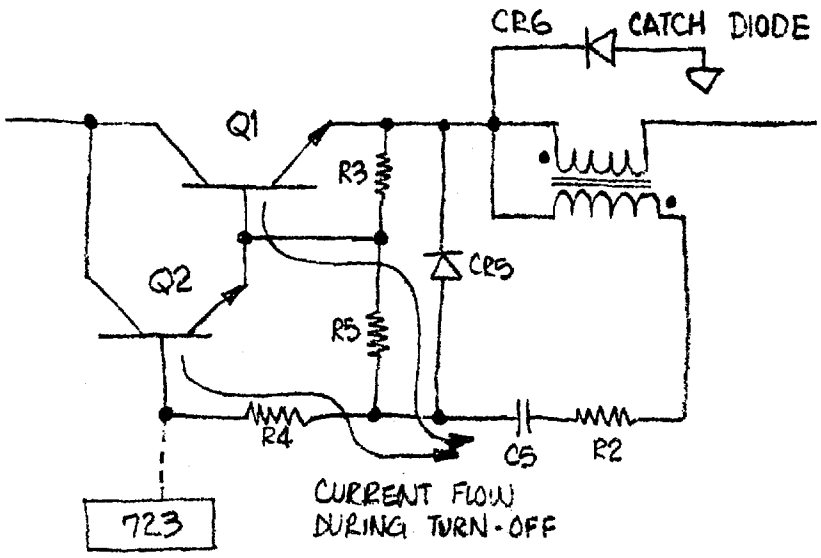


Figure 5
Fast Turn Off Circuit

One way to minimize losses in the switch transistor is to turn it off quickly. This circuit is designed to speed transistor turn off by depleting charge carriers from the base region of Q1 and Q2.

When the 723 tells Q1 to turn off, the dotted end of the upper (as shown here) winding of L2 goes essentially to ground, and the dotted end of the lower winding goes negative by about 6V. This negative voltage applied to the bases of Q1 and Q2 helps deplete charge carriers. The current flows thru R5, R4, C5 and R2 during this time. CR5 discharges C5 when Q1 turns on.

SATURATING REACTOR INVERTER

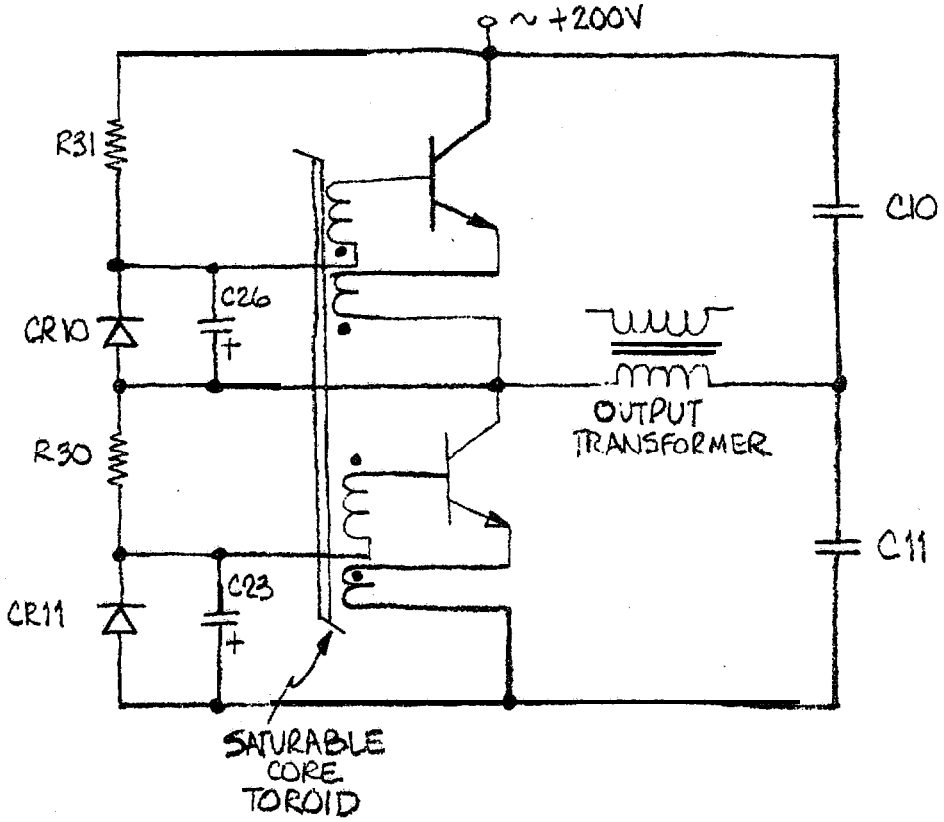


Figure 6
Saturating Reactor Inverter

The saturating reactor inverter is a free-running square wave generator. Its frequency is determined by the time taken to saturate a small ferrite toroid, which also functions as a current feedback transformer to drive the bases of Q10 and Q11. The inverter shown is a half bridge. The push-pull inverter works in an analogous way.

When power is applied to the inverter, leakage current through R31 will begin to turn on Q10. The undotted side of the output transformer will begin to become positive. Current through Q10's emitter winding further biases Q10 into saturation and Q11 off. Current builds up in Q10's emitter winding and in the output transformer until the saturable core saturates.

At that point, the small, saturating core can contain no more flux. The transformer action in the saturable core diminishes, and hence the base drive to Q10 from its emitter winding virtually stops. Q10 begins to come out of saturation (because emitter current is rising while base drive is decreasing). Emitter current now begins to decrease. This implies that di/dt has reversed, and the saturating core now reverses polarity. Qll begins to be driven on while Q10 is cut off. In practice, the transfer of current from one transistor to the other is "snap action" rapid.

Capacitors C10 and C11 define a voltage midway between the +200V supply and ground. This allows the primary of the transformer not connected to the capacitors to swing symmetrically between the supply voltage and ground. The capacitors also absorb the primary transformer current.

MAIN FEEDBACK LOOP

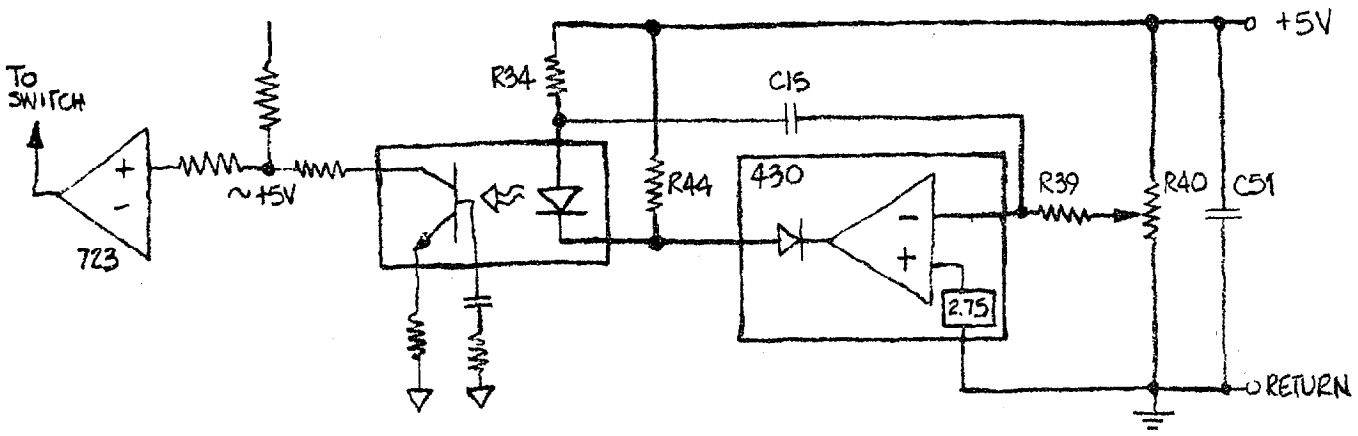


Figure 7
Main Feedback Loop

This circuit maintains the +5V output at a constant voltage. The auxilliary (other than +5V) outputs are semi-regulated. That is, because of the tight magnetic coupling of all the outputs, a change in loading on any auxilliary output affects the magnetic flux of the transformer, which tends to change the +5V output. The feedback loop compensates for this minute change in the +5V output and in so doing partially compensates for the load change on the auxilliary output. Boschert calls this "semi-regulation."

The loop compares a voltage divided down from the +5V output via R40 to a 2.75V reference within the 430 integrated circuit. A proportional current is sunk by the 430, driving the opto-isolator diode via R34. R44 provides bias current for the 430.

The opto-isolator varies the voltage on the non-inverting input of the switch control circuit (the 723). This has the effect of varying the pulse width on the switch. The pulse width ultimately controls the output voltages by varying the voltage to the primary of the output transformer.

C15 frequency compensates the 430. The bandwidth of this circuit is 1-2kHz.

The use of an opto-isolator is necessary to isolate the primary side of the supply, which floats at the input line potential, from the secondary, which is often grounded to the system chassis.

SHORT CIRCUIT PROTECTION

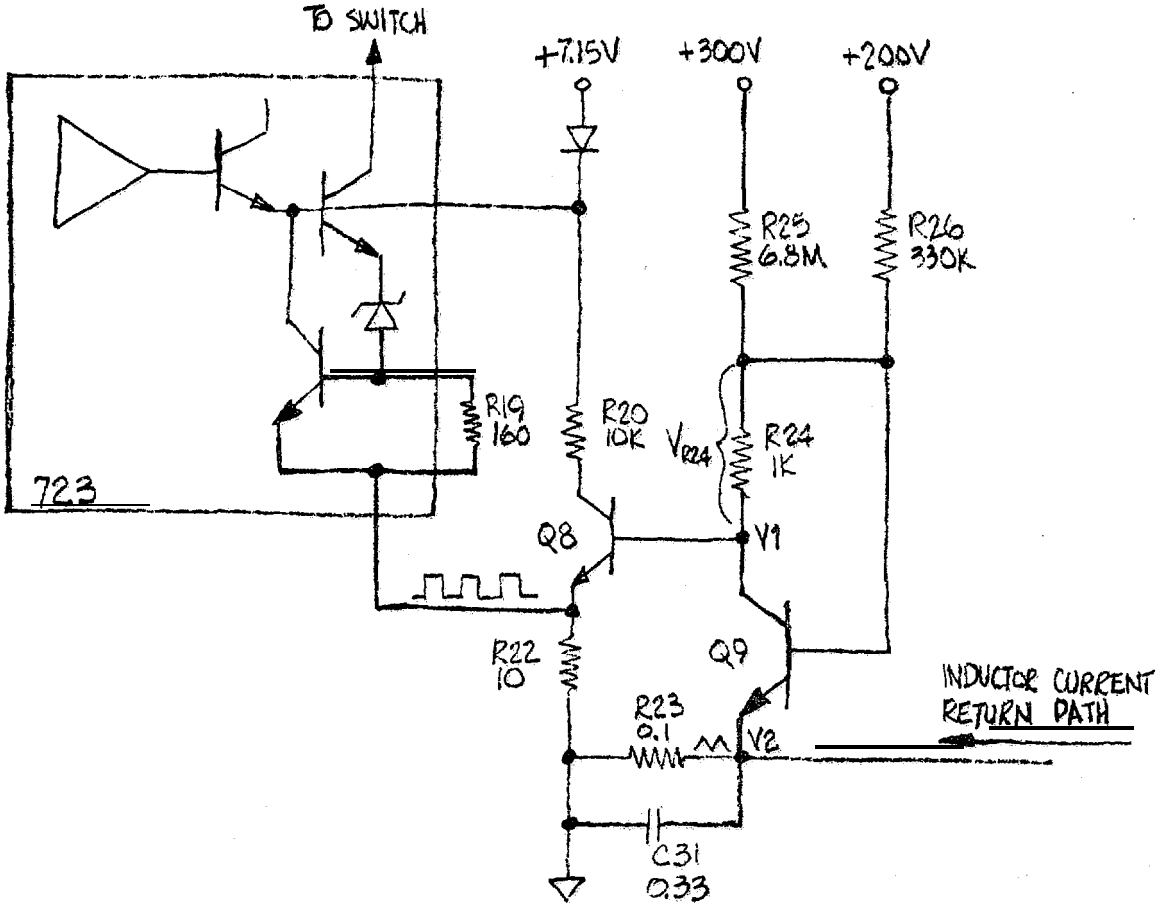


Figure 8
Short Circuit Protection
(Foldback Circuit)

The foldback circuit performs two basic functions. It senses the regulated voltage to the inverter, and it senses the primary current through the inverter. These two voltages are electronically added together, and when their sum exceed a prespecified value, foldback begins. In normal operation, the foldback circuit senses the power (voltage and current) in the primary circuit which, minus losses, is being delivered to the outputs. If the secondary tries to deliver too much power, as in the case of a short, the circuit will sense this and fold back all outputs to a very low voltage.

Q9 performs the combined functions of voltage and current sensing. It operates in an almost saturated state, and the output is the collector voltage, V1. This voltage increases with decreasing (+200V nominal) regulated buss voltage and increasing primary inductor current. To understand why, it is helpful to describe V1 as:

$$V1 = V2 + 650mV - V_{R\ 24}$$

Notice that V2 increases with increasing primary current (the current from L2) and V_{R24} decreases with decreasing (+200V nominal) regulated buss voltage. Also note that V2 ramps up during each cycle, which affects Q8 as described next.

Q8 controls the foldback function. Under normal conditions, Q8 conducts very little during each cycle, which does not affect the operation of the 723. However, as V1 rises higher during each cycle, Q8 conducts more and more base drive from the output stage of the 723. Eventually, when V1 rises high enough during a cycle, Q8 absorbs all the base drive and shuts off the switch prematurely. The higher V1 rises, the more prematurely the switch gets shut off. This, of course, means the duty cycle of the switch (Ql) is changing, reducing the (+200V nominal) regulated buss voltage. As the buss voltage drops, V1 rises higher yet and the switch is shut off even more prematurely. This is the essence of the foldback function.

OVERVOLTAGE PROTECTION

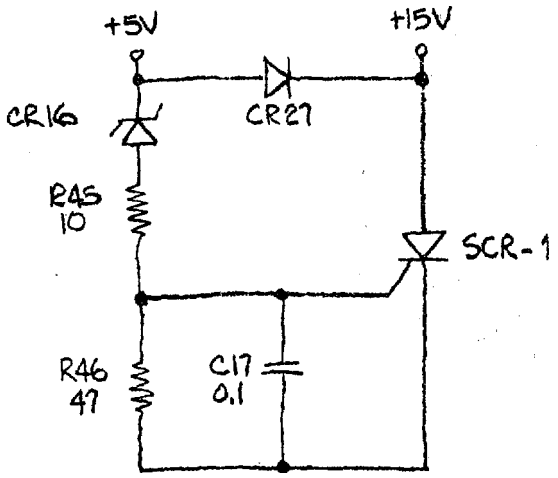


Figure 9
Overvoltage Circuit

The purpose of overvoltage protection is to protect the user's load from an overvoltage condition caused by supply failure. All standard Boschert supplies incorporate this feature. Also, Boschert supplies require OVP on only one output. In most cases, all outputs have simple passive filtering, and if one output should rise due to supply failure, they all would rise, including the +5V output. Thus, protection is needed on only one output.

If the +5V output exceeds 6.25V $\pm$ 0.75V, the SCR fires and pulls the +15V supply down (to about 1.5V), activating the short circuit protection. The +15V supply is used to insure clean foldback. Pulling the +5V down to 1.5V might not exceed the foldback point, and might simply deliver a lot of power to the SCR, destroying it.

This circuit can be cleared by cycling line power.

NEGATIVE RESISTOR CIRCUIT

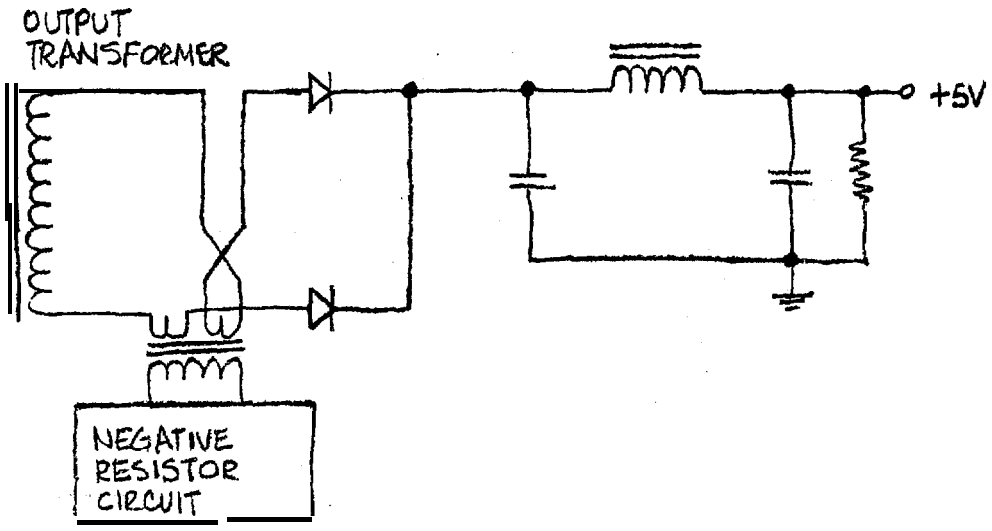


Figure 10
Negative Resistor Circuit

The negative resistor circuit is an active circuit which compensates for IR drops in the +5V line caused by high output current. This magnetically coupled circuit adds a few tenths of a volt drop to the output line at low output current which effectively bucks the output voltage. At high currents the voltage drop is about zero. This voltage/current characteristic is opposite that of an ordinary resistor, hence the name "negative resistor."

The net effect of this circuit is that the +5V output voltage, even without the main feedback loop to regulate it, would change very little with large changes in load current. Since the main feedback loop does not have to compensate for large changes in +5V load, it also does not radically affect the auxilliary output voltages.

OVER POWER CIRCUIT BREAKER

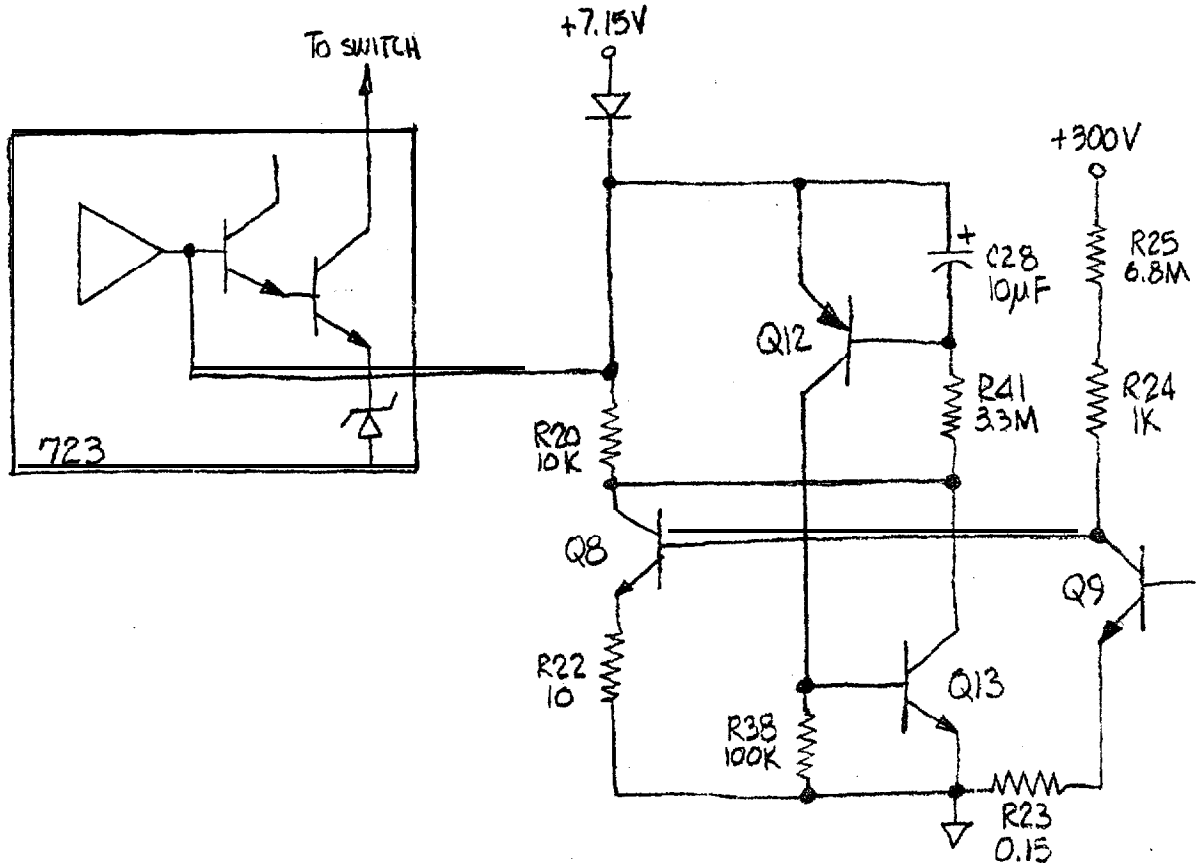


Figure 11
Over Power Circuit Breaker

The purpose of this circuit is to prevent the overheating of some power supplies if left shorted or crowbarred for an extended period of time.

The circuit consists of **Q12,** Q13, C28, R41 and R38. **Q12** and **Q13** form a regenerative pair, or "SCR" in effect. If one transistor turns on and saturates, the other will turn on also, and sustain this mutually saturated condition.

At the onset of foldback, Q8 will saturate for an instant during each cycle. The deeper the foldback, the longer Q8 stays saturated during each cycle. With a dead short on an output, Q8 will remain saturated for a good portion of each cycle.

Whenever Q8 saturates, C28 is allowed to charge thru R41. During normal operation, C28 remains discharged by R41 and R20, When the supply goes into foldback, Q8 remains saturated long enough during each cycle to begin to charge C28. Eventually C28 forward biases Q12, which fires Q13, and the supply completely stops. The time taken to turn on Q12 varies from about 30 sec at the point where foldback begins to about 5 sec in full foldback (output shorted.) This circuit can be cleared by interrupting line power for about 60 sec.

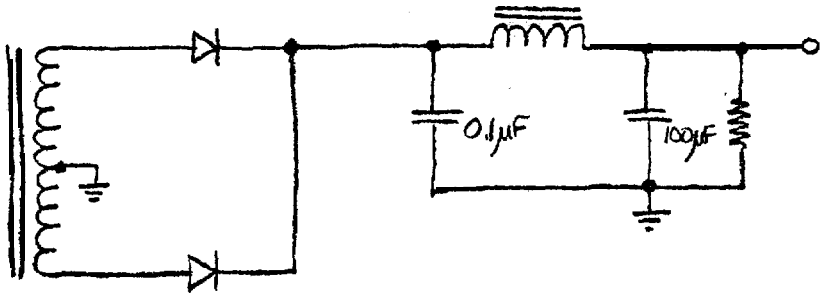


Figure 12
Typical Output

Practically all outputs on Boschert supplies are full wave rectified, then passively filtered using either an L or PI section filter. This provides quite good regulation at a minimum of cost. Almost all supply outputs share a common return. The bleeder resistor on the output helps discharge the output capacitor as well as providing a minimum loading on the power supply.

Some outputs are fused or have the ability to accept a three terminal regulator. See data sheets for further details.

TWO STAGE (PUSH-PULL) POWER SUPPLY

TROUBLESHOOTING INSTRUCTIONS

EQUIPMENT NEEDED:

- 1 50 MHz oscilloscope with isolated ground
- 1 DVM or VTVM with isolated ground
- 1 200 ohms, 200W variable resistor variable load station or several load resistors for each output
- 1 10A Full Scale Ammeter
- 1 10A Variac (0-130VAC)
- 1 0-130 VAC AC Voltmeter

NOTE: It is desirable to use an AC isolation transformer in performing all tests to minimize shock hazard. The kVA rating should be at least 3 times the supply power to avoid waveform distortion.

CAUTION: Lethal voltages are present in this supply. Only authorized service technicians should perform these tests. Use AC isolated equipment in all tests.

1. Open the major feedback loop by locating the yellow jumper wire on the p.c. board and disconnect it. This opens the circuit between the primary inductor and inverter, facilitating repair. Connect a 200 ohm, 200W variable resistor in series with a 10A F.S. Ammeter between the pin sticking up on the p.c. board that the yellow wire was connected to, and the junction of Q9 emitter and R23.

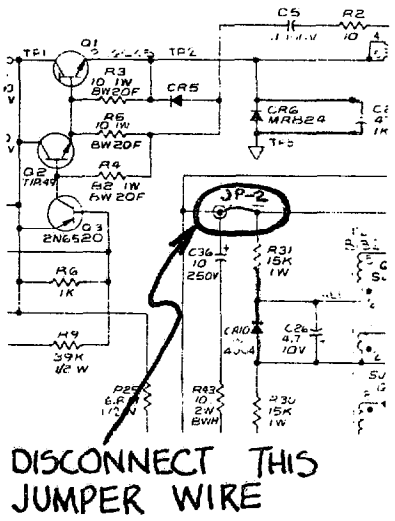


Figure 13

2. With the supply disconnected from the line, use the DVM to check the continuity of the input fuse (F1), the input thermistor (R1) and the input bridge diodes (CR1 thru CR4), for Proper condition.

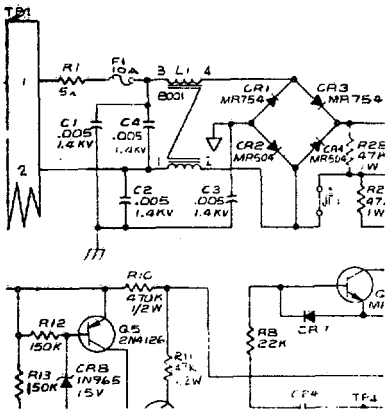


Figure 14

3. Use the DVM to check the two input terminals for shorts using the lowest ohm scale. Probe the two AC input terminals for any short.

4. unplug the opto-isolator and insert a 10K 1/4W resistor between the sockets where pins 4 and 5 of the opto-isolator were. If the opto-isolator is soldered in, clip or solder the 10K resistor between pins 4 and 5 of the opto-isolator. (Be sure to carefully note the position of the 6 pin mini-dip in its 8 pin socket, so that it can be properly replaced.)

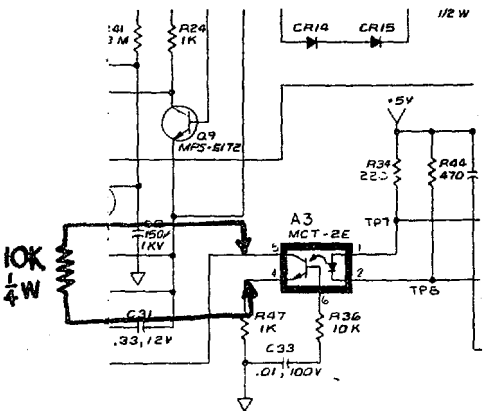


Figure 15

5. Look at the voltage across the catch diode CR6, which is the emitter voltage of Ql. Connect the supply to the Variac, and slowly ramp up the input voltage while watching the DVM. Do not exceed 30VAC RMS on the input. Does the voltage across CR6 ramp up as you increase the input voltage?

Yes - Go to Step 6

No - Go to Step 9

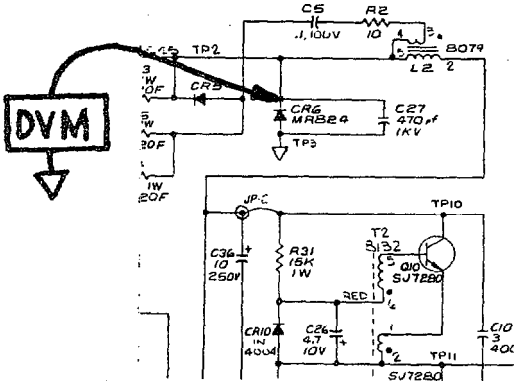


Figure 16

6. Use a DVM or lab supply set for 5V, current limited at 100mA, to check the continuity of Ql, Q2 and Q3 for shorts. Are there any shorted transistors?

   Yes - Replace appropriate transistors, and repeat Step 5

   No - Go to Step 7

7. Set the AC input voltage to 30VAC RMS with the Variac. Check the voltage across R9.

   0 Volts - The problem lies with Ql, Q2 or Q3. Check more carefully for a current leakage path that might turn on Ql. Repair, and go to Step 5.

   Other than 0 Volts - Go to Step 8.

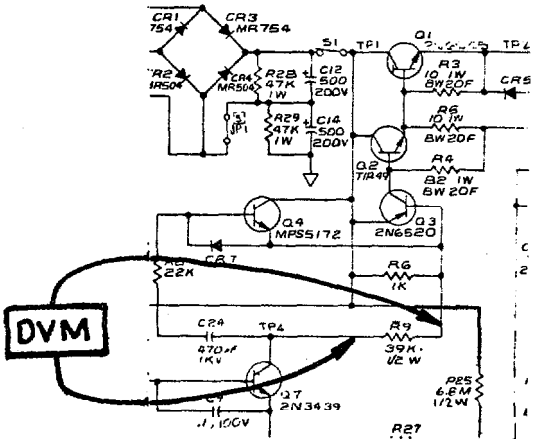


Figure 17

8. With the AC input voltage still at 30VAC RMS, look at the voltage at Pin 13 of A1.

V ≤ 7.0V Check Q7 for shorts and replace if necessary. Go to Step 5.

V > 7.0V Replace A1, and look for any problems around A1. Go to step 5.

NOTE: The following tests should be performed for a few seconds only to prevent overheating the 200 ohm variable resistor, with the input set to the nominal line input voltage.

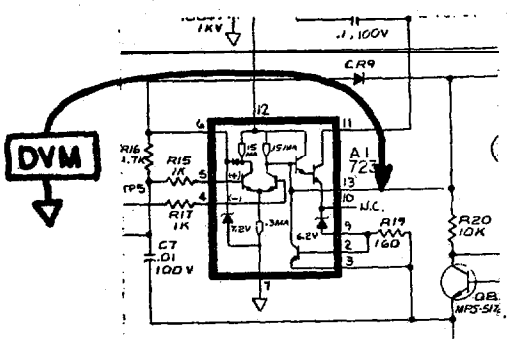


Figure 18

9. Use the DVM to check the voltage across the input capacitors. Is the voltage as follows?

115VAC only supply

V = **150V ±10V**

220VAC only supply

115VAC/220VAC strappable supply

v = 3oov **±20V**

Yes - Go to Step 10

No - Look for faulty component in input section - repeat Step 9.

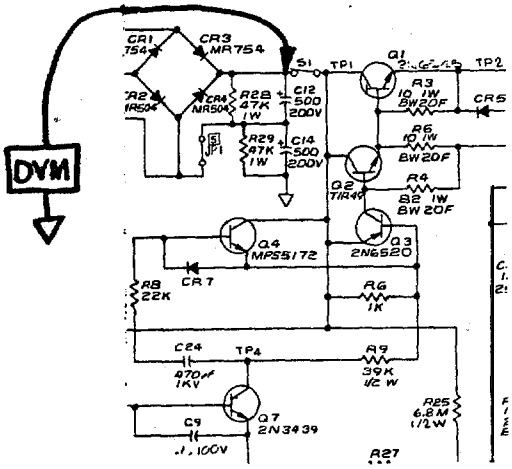


Figure 19

10. Check the voltage across the catch diode CR6 for the proper waveform.

Good waveform - Go to Step 16
Bad Waveform - Go to Step 11

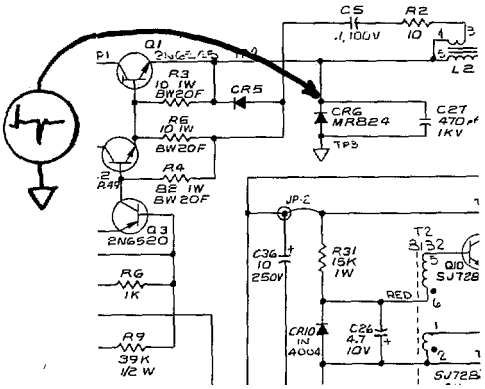


Figure 20

Proper waveform for 115VAC only supplies. 220VAC only or 115VAC/220VAC strappable supplies are double the amplitude.

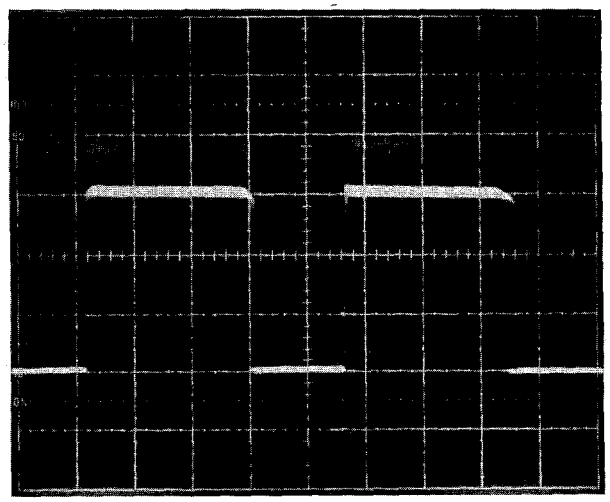


Figure 21
Horizontal - 10 μs/Div.
Vertical - 50V/Div.

11. Check the voltage on Pin 12 of A1 (The 723 which controls the switch) with respect to Pin 7 (Primary ground). Is this voltage +15V ±1V?

Yes - Go to Step 14
No - Go to Step 12

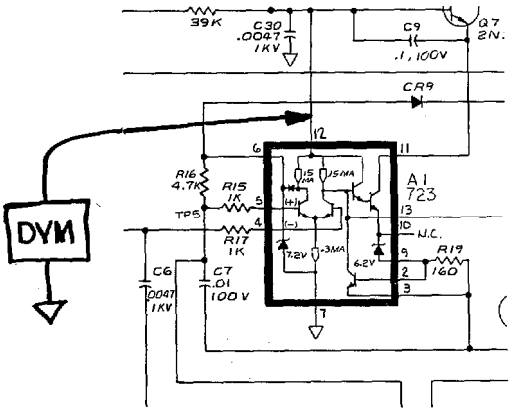


Figure 22

12. Check the **+15V** linear regulator. Is the voltage across the **zener (CR8)** **+15V +1V?**

Yes - Go to Step 13

No - Replace Zener, go to Step **10**

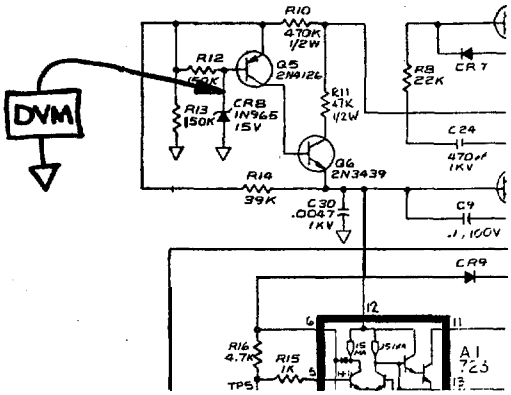


Figure 23

13. Unsolder Pin 12 of Al, the 723 which controls the switch. (A "solder-sucker" will remove all the solder from pin 12, so that it doesn't make contact with the PCB trace. This is much easier than actually talking the 723 out.) Is the emitter voltage of Q6 = **+15V ±1V?**

Yes - **Replace** Al, go to Step 10

No - Check **Q5** and Q6, replace as necessary. Go to Step 10

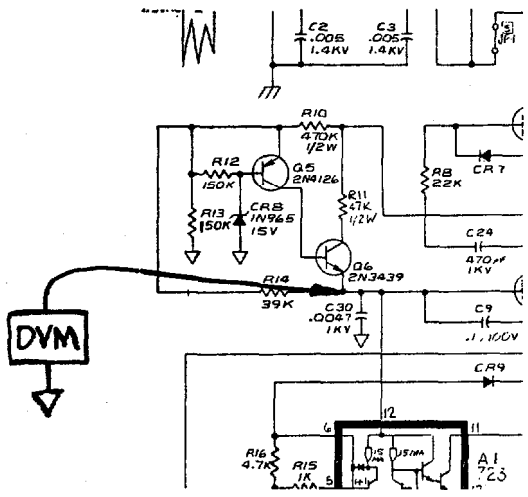


Figure 24

14. Check the following voltages on A7 (the 723 which controls the switch) with respect to Pin 7, primary ground.

Pin 5 +5V ±0.5V
Pin 6 +7V ±0.5V
Pin 13 +8V ±0.5V

Pin 5 bad - Check R15, R16, R33, R47 and C7

Pin 6 bad - Replace Al, repeat Step 14

Pin 13 bad - Go to Step 15

All good - Check drive string continuity, including R6, R9, Q7, C9, R19 and R22.

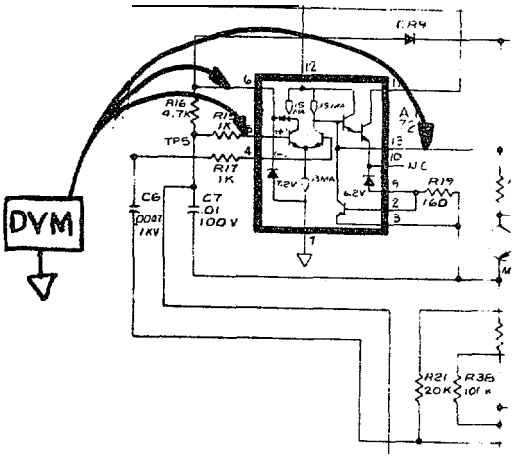


Figure 25

15. Check the voltage across R20. Is it 0 volts?

Yes - Check drive string continuity, including R6, R9, Q7, C9, R19 and R22. Go to Step 10.

No - Replace Q8 and Q9, go to Step 10

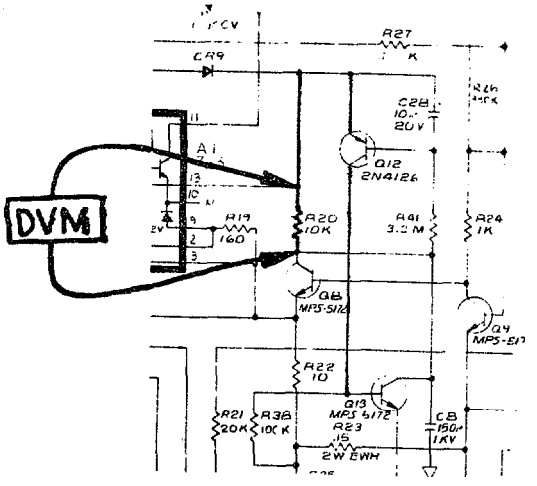


Figure 26

16. Check the primary power foldback circuit. Turn on the supply to 115VAC with the variac. Increase the primary current by reducing the resistance of the 200 ohm variable resistor. Does the current reach a maximum then foldback according to Table II?

Yes - Go to Step 19

No - If the supply doesn't foldback, Q1 will be destroyed. Replace it before proceeding. Start with Step 10 and work thru to Step 17. SKIP STEP 16.

MODEL	IMAX (Amps)	IFOLDBACK (Amps)
OL130	0.9-1.0	0.2-0.7
OL150*	4.0-5.0	0.8-2.0
OL151*	4.0-5.0	0.8-2.0
OL152*	4.0-5.0	0.8-2.0
OL153*	4.0-5.0	0.8-2.0
OL200	1.5-2.5	0.1-1.0
OL301	3.4-4.6	1.0-2.3
OL400	3.3-3.8	0.2-1.2

TABLE II

*Divide IMAX and IFOLDBACK limits by 2 for 220VAC input supplies.

17. Check the operation of Q9. Turn on the supply and set the 200 ohm resistor to 1A. Check the collector voltage of Q9 with respect to primary ground with an oscilloscope. Is it 0.1V ±0.05V with a 50mV triangle wave on it, as indicated in Figure 28?

Yes - Go to Step 18

No - Replace Q9, go to Step 18

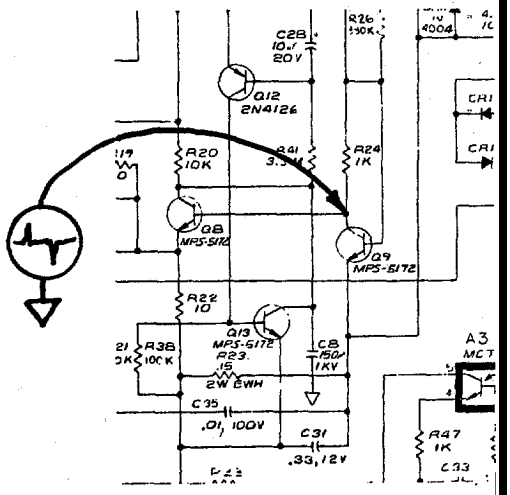


Figure 27

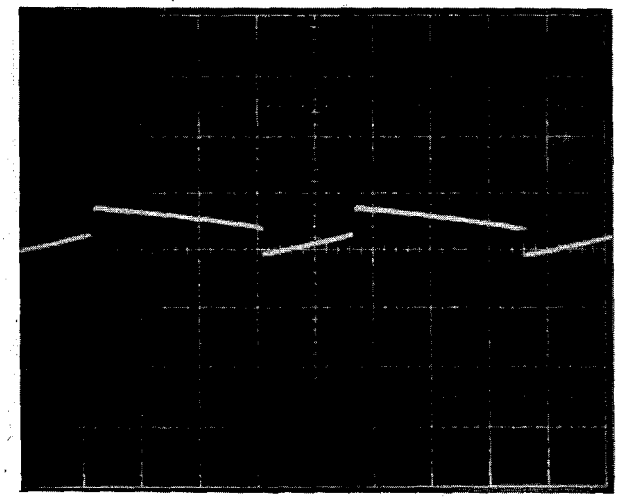


Figure 28

Horizontal - 10 µs/Div.

Vertical - 50mV/Div.

78. Check the operation of Q8. Watch the switch waveforem across catch diode CR6. Now short R24 with a jumper.

NOTE: Remember that R24 floats at the line voltage--don't touch it with your hands.

Does the duty cycle change to a very low duty cycle, as shown in Figures 31 and 32?

Yes - Replace R26, go to Step 16

NO - Replace Q8, go to Step 16

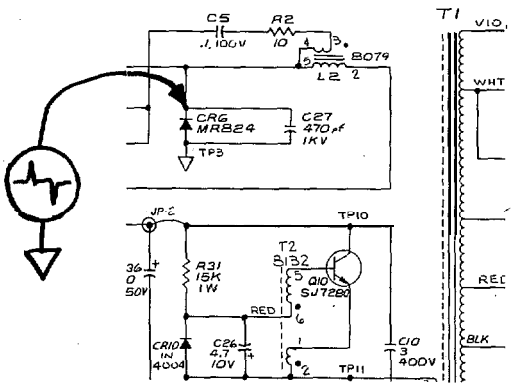


Figure 29

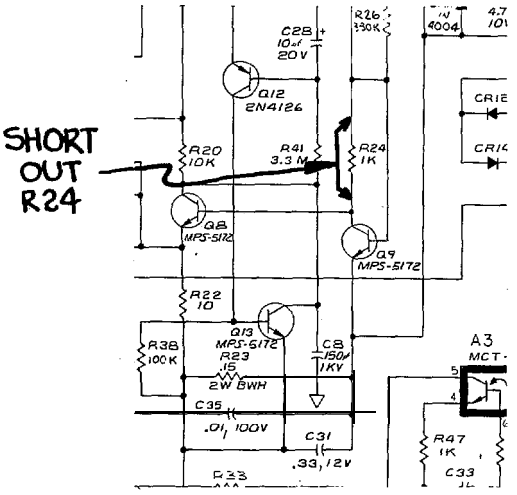


Figure 30

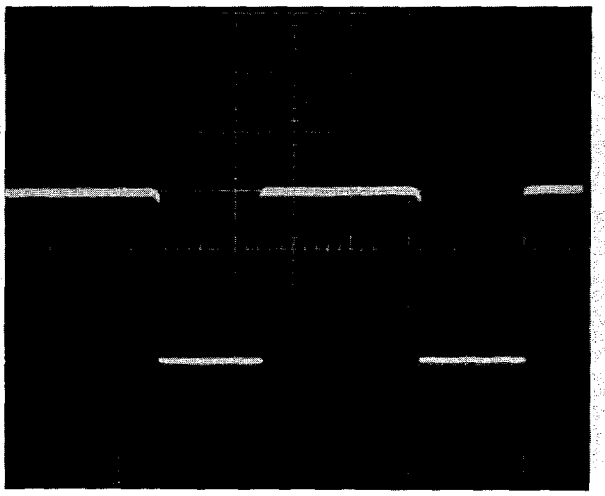


Figure 31 - R24 not shorted
Horizontal - 10 us/Div.
Vertical - 50V/Div.

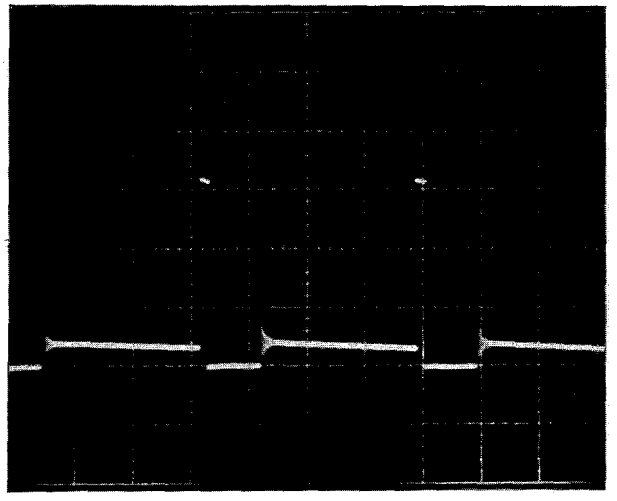


Figure 32 - R24 Shorted
Horizontal - 10 us/Div.
Vertical - 50V/Div.

19. Turn off the supply. Disconnect the 200 ohm variable resistor and 10A meter. Reconnect the yellow flying lead described in Step 1. Disconnect the 10K resistor across pins 3 and 4 of the opto-isolator, and replace the opto-isolator.

20. Turn on the supply with 50% load on all outputs. Check the inverter waveform for the proper amplitude and waveshape.

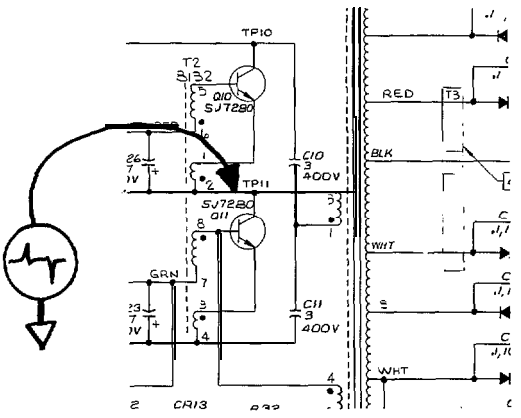


Figure 33

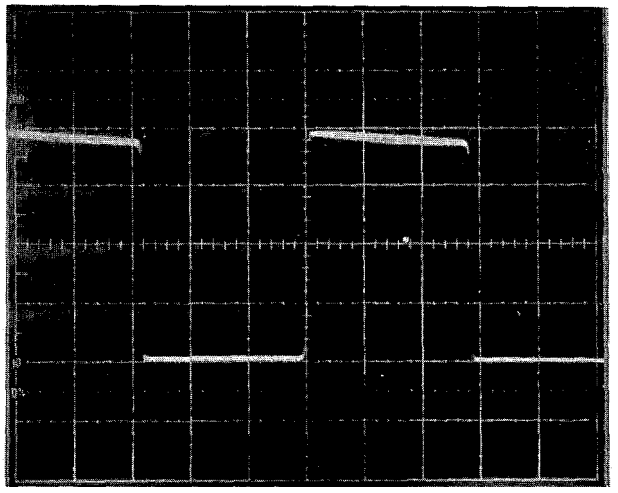


Figure 34

Proper Waveform

Horizontal - 10 us/Div.
Vertical - 50V/Div.

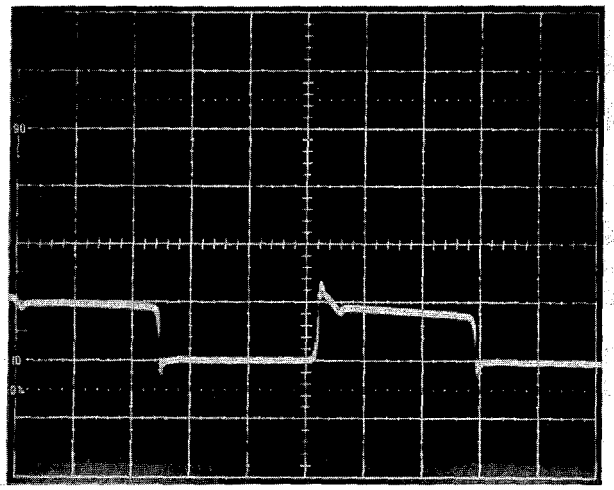


Figure 35 - OVP Tripped
Horizontal - 10 μs/Div.
Vertical - 50V/Div.

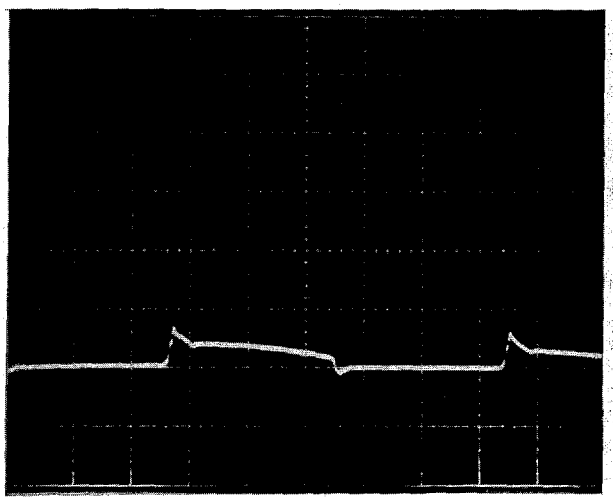


Figure 36 - Output Diode Shorted
Horizontal - 10 μs/Div.
Vertical - 50V/Div.

21. Check the +5V output for ripple and noise. Is the total ripple and noise less than 100 mV P-P?

Yes - Go to Step 22

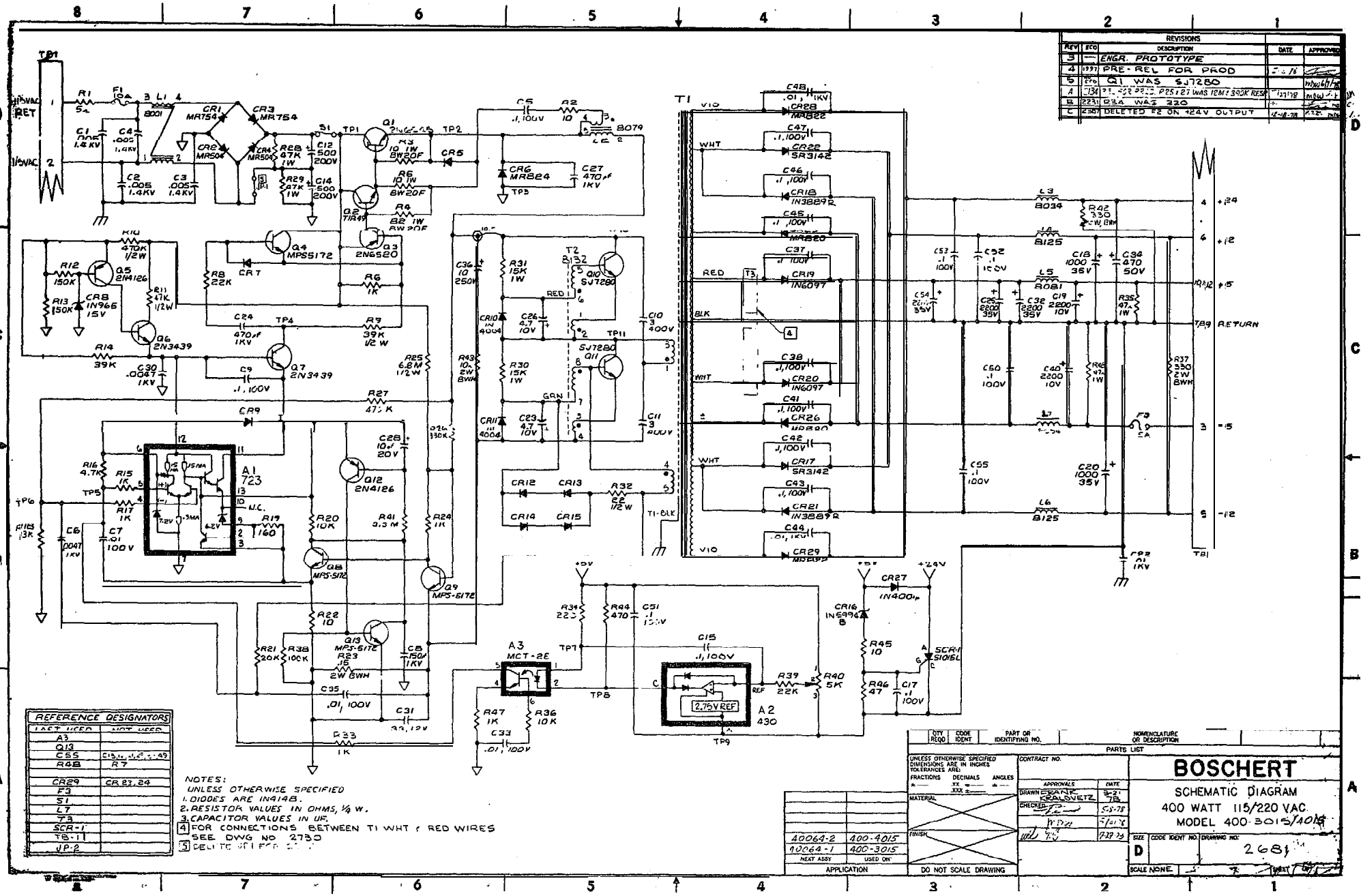
NO - Look at output caps, solder joints, etc. If there is a 1-5kHz frequency present replace A3 and repeat Step 21.

22. Check the +5V output regulation. Does it vary less than 50mV as the load is adjusted from 20% to 100% max rated load?

Yes - Go to Step 23

NO - Replace A3, the opto-isolator, and Repeat Step 22

23. Check all outputs for proper regulation. If any output voltage falls dramatically with increasing load, check for open rectifier in output circuit.


REVISIONS
REV | ECO | DESCRIPTION | DATE | APPROVED
3 | — | ENGR. PROTOTYPE
4 | 1997 | PRE-REL FOR PROD
5 | 870 | Q1 WAS SJ7280
A | 134 | R2 .05Ω 225 W WAS 12M, 300K RESP
B | 2231 | D34 WAS 320
C | 2367 | DELETED F2 ON +24V OUTPUT
REFERENCE DESIGNATORS
LAST USED | NEXT USED
A3
Q13
C55 | C56, 57, 58, 59
R48 | R7
CR29 | CR 83, 84
F3
S1
L7
T3
SCR-1
TB-11
JP-2
NOTES:
UNLESS OTHERWISE SPECIFIED
1. DIODES ARE IN4148.
2. RESISTOR VALUES IN OHMS, ¼ W.
3. CAPACITOR VALUES IN UF.
4 FOR CONNECTIONS BETWEEN T1 WHT & RED WIRES
SEE DWG NO 2730
5 DELETE IF F2 IS
PARTS LIST
QTY REQD | CODE IDENT | PART OR IDENTIFYING NO. | NOMENCLATURE OR DESCRIPTION
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES ARE:
FRACTIONS ± | DECIMALS .XX ± .XXX ± | ANGLES ±
MATERIAL
FINISH
NEXT ASSY | USED ON
40064-2 | 400-4015
40064-1 | 400-3015
APPLICATION
DO NOT SCALE DRAWING
CONTRACT NO.
APPROVALS | DATE
DRAWN FRANK KRALOVETZ 3-21-78
CHECKED 5-8-78
5/11/78
7-27-78
BOSCHERT
SCHEMATIC DIAGRAM
400 WATT 115/220 VAC
MODEL 400-3015/4015
SIZE D | CODE IDENT NO. | DRAWING NO. 2681
SCALE NONE | SHEET

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

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